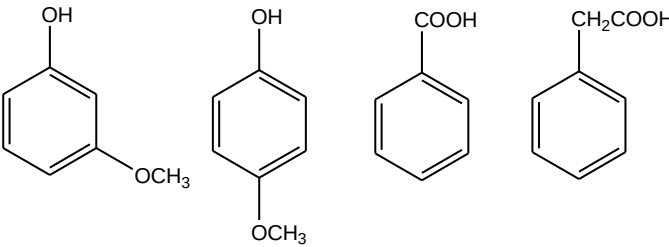
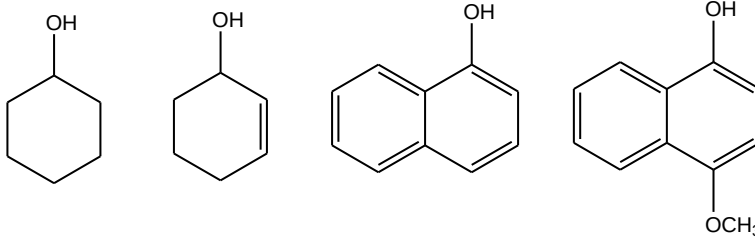
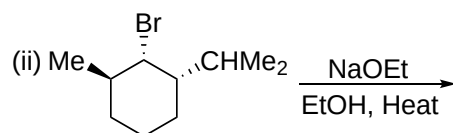
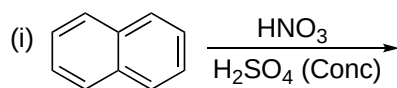


**ACML 101**  
**Organic Chemistry Assignment 2**

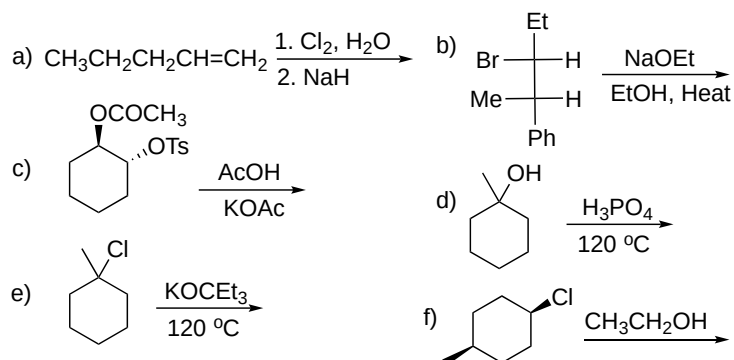
Q1. Arrange the following in increasing order of their relative acidity.

- a) 
- b) 
- c)      HOAc              HOCH<sub>3</sub>              HCN              HOTs
- d)      CH<sub>3</sub>COOH      NO<sub>2</sub>CH<sub>2</sub>COOH      HOOC-CH<sub>2</sub>-COOH      HOOC-CH<sub>2</sub>-COO<sup>-</sup>
- e)      O<sub>2</sub>N-CH<sub>2</sub>-COOH      H<sub>3</sub>COCH<sub>2</sub>COOH      CH<sub>3</sub>CH<sub>2</sub>COOH      Me<sub>3</sub>N<sup>+</sup>-CH<sub>2</sub>COOH

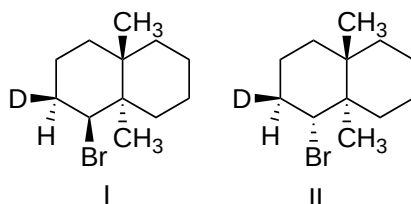
Q2. Which of the following reactions will show kinetic hydrogen isotope effect? Explain with the help of mechanism and identify the RDS. Give the product formed in reaction (ii) and draw the labeled energy profile diagram.



Q3. Identify the major product in the following reactions with correct stereochemistry wherever applicable.



Q4. On attempted  $\text{E}_2$  reaction, (I) reacted much faster than (II). Write the product of the reaction in each case and explain the reason.

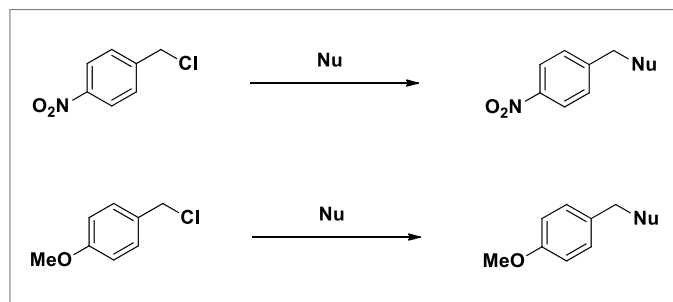


Q5. For the  $\text{S}_\text{N}2$  reaction,

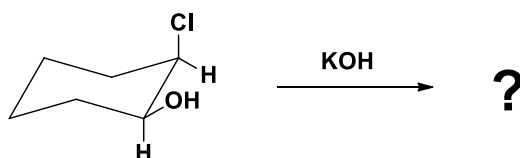
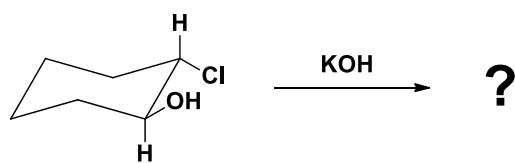


- What is the rate expression for the reaction?
- Draw an energy diagram for the reaction. Label all parts. You may assume that the products are of lower energy than the starting reagents.
- What will be the effect on the rate of reaction if the concentration of *n*-butyl bromide is doubled?
- What will be the effect on the rate of reaction if the concentration of the NaOH is halved?
- How will the rate of the  $\text{S}_\text{N}2$  reaction change if the solvent is changed to 80% ethanol and 20% water?

Q6. Predict the reaction mechanism and compare the rates in two different sets of reactions given below:



Q7. Predict the product(s) structure in the following two sets of reactions.



Q8. When the deuterium labelled compound shown below is subjected to dehydrohalogenation using alcoholic base, the only alkene product is 3-methylcyclohexene with no deuterium.

